Allan Zhang

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Education

University of California, Los Angeles *BS in Computational Mathematics* *Sept 2024 – June 2028*

- **Relevant Coursework:** Multivariable Calculus, Linear Algebra, Differential Equations, Discrete Structures, Probability Theory, Data Structures and Algorithms, Large Scale Machine Learning (graduate coursework)

Experience

Research Intern | DatologyAI *Jan 2026 - April 2026*

- Developed a synthetic data generation pipeline producing 3.4M multi-turn reasoning traces for VLM pretraining, with intermediate steps grounded in visual evidence (bounding boxes, extracted values, structural hierarchies)
- Built a reinforcement learning framework for VLMs from scratch to generate and evaluate high-quality reasoning traces; optimized weight synchronization between inference workers and trainers
- Trained custom 4B VLMs on new pretraining mixtures incorporating reasoning data, improving RealWorldQA from 49.9% to 53.5% and average benchmark score from 58.7% to 62.0% over the internal curation baseline

Undergraduate Research Assistant | BigML @UCLA *Nov 2024 – Present*

- Conducting research on data-efficient training for LLMs, focusing on data selection for reinforcement learning and synthetic data generation. Advised by Prof. Baharan Mirzasoleiman
- Co-designed a data-efficient variant of GRPO, achieving 30% faster convergence and a 40% reduction in required training samples relative to vanilla GRPO

Projects

Torchure (Pure PyTorch Training Stack) *torchure*

- Built a distributed LLM pretraining stack in pure PyTorch: implemented Qwen3, streaming dataloader with sequence packing, CUDA prefetch, and stateful checkpointing
- Lifted single-GPU throughput for Qwen3-0.6B by 52% (35.8k → 54.5k tok/s) and cut peak memory 20% with per-block torch.compile and a custom fused linear + cross-entropy kernel
- Implemented DDP and FSDP from scratch: bucketed gradient all-reduce overlapped with backward via post-accumulation hooks, achieved 198k tok/s on 4×H100 at 92% scaling efficiency

Thinking Constrained GRPO *grpo-thinking-budget*

- Implemented constrained reasoning for LLMs by enforcing thinking token budgets during GRPO training, enabling fine-grained control over model inference costs
- Designed a two-phase generation system with vLLM that separates thinking and response phases, with automatic delimiter injection when thinking budget is exhausted
- Created fast inference pipeline using vLLM with two-phase generation to control thinking vs. response token allocation for evaluations

Publications/Preprints

Beyond What Seems Necessary: Hidden Gains from Scaling Training-Time Reasoning Length under Outcome Supervision *NeurIPS 2026*
Yihao Xue, Allan Zhang, Jianhao Huang, Amit Sahai, Baharan Mirzasoleiman

Understanding the Role of Reasoning Data in Pretraining *Under Review ICLR 2027*
Jianhao Huang, Allan Zhang, Baharan Mirzasoleiman

Skills

Programming Languages and Frameworks: Python, PyTorch, JAX, NumPy, Matplotlib, TensorFlow, scikit-learn, OpenCV, Hugging Face, LaTeX, C++, Bash, vLLM, SGLang